Name: ____ **NetID:** ______________________

ECON 1001 — Spring 2026 — Midterm 2

Version C

All answers must go on the separate response sheet. Each problem or subproblem is worth 4 points.

1. Typically to celebrate Diwali, many people light fireworks. But last weekend, the Indian city of Delhi prohibited people from lighting fireworks because of their impact on air quality. Assuming the Delhi government makes decisions based solely on maximizing efficiency, evidently the Delhi government

A) Underestimated how much people enjoy fireworks.

B) Believed that clean air was a rival good.

C) Believed that the total social cost of the fireworks was greater than the total social benefit of the fireworks.

D) Believed that the cost to people who purchased the fireworks was smaller than the positive externality from the fireworks.

2. Without government provision of public goods, public goods tend to be underproduced relative to the efficient quantity, because ________________.

A) Individuals cannot be relied on to act in their own best interests.

B) The options labeled (⋄) and (■) are both correct.

C) Even if the cost of the public good is less than an individual's private benefit from the good, the individual may not purchase the good because they think someone else will. (■)

D) The cost of the public good is greater than most people's individual marginal private benefit from the good, so most individuals don't buy the good. (⋄)

3. Suppose Frinkle works in a coffee shop (assume leisure is a normal good for Frinkle, as usual). Business has been bad, so her boss lowers her wage. That causes her to work even more. Evidently, for Frinkle:

A) Both the substitution effect and the income effect of the wage change push her to work more.

B) Her individual labor supply curve is backward-bending.

C) The substitution effect of the wage change is stronger than the income effect.

D) The income effect of the wage change is stronger than the substitution effect.

4. Consider the *market* for low-skilled labor. When the hourly wage increases by $2 an hour, *ceteris paribus*, you expect

A) The total amount of low-skilled labor supplied to decrease, because of the income effect.

B) The total amount of low-skilled labor supplied to increase from both the extensive and intensive margins.

C) The total amount of low-skilled labor to stay the same, because of the opposing income and substitution effects.

D) A backward-bending labor supply curve.

5. Suppose that (1) you are the manager of a widget factory, (2) the market for widgets is perfectly competitive, and (3) the market for labor is perfectly competitive. You developed an app that allows each of your factory workers to make widgets faster. As a result,

A) You lower the price you charge for widgets.

B) Your demand for labor decreases because you need fewer workers to make the same number of widgets.

C) You raise the wage you pay workers.

D) You hire more workers.

6. A new law requires large employers like Big Store Co. to provide more generous family leave benefits to their employees. That is, if an employee needs to provide care for a newborn or a sick family member, the firm must give them some paid time off. This law will lead to:

A) An increase in the amount of labor Big Store Co. employs.

B) An increase in the MRPL (Marginal Revenue Product of Labor).

C) Not enough information to choose another option.

D) A decrease in the amount of labor Big Store Co. employs.

7. Suppose there is a pandemic which causes many people to avoid buses and pushes up the price of an Uber ride. Also, the pandemic causes many Uber drivers to either quit or require higher wages. As a consequence of those changes, our model leads us to expect

A) None of the other options.

B) A definite increase in the wage of an Uber driver but an ambiguous effect on the number of Uber drivers.

C) A definite increase in the wage of an Uber driver and an increase in the number of Uber drivers.

D) A definite increase in the wage of an Uber driver and a fall in the number of Uber drivers.

8. Suppose a hiring manager decides to hire a man instead of a young, married woman because they think the woman is more likely to have a baby and take parental leave (paid time off from work) in the near future. This is an example of:

A) Choosing an individual based on the individual's demonstrated productivity.

B) Implicit bias.

C) Prejudice.

D) Statistical discrimination.

9. Imagine it's Friday and you want to go to The Club. Ouch, Ubers (or taxis) cost \$30. You ask a friend, who was splitting the cost of a \$30 Uber with her boyfriend, if you can join them. In order for the total gains from sharing the ride to be split equally among the three riders, what should each person pay?

A) None of the other options.

B) You pay \$20, your friend pays \$5, and her boyfriend pays \$5.

C) You each pay \$10.

D) You pay \$15, your friend pays \$7.50, and her boyfriend pays \$7.50.

10. Consider Figure 1 (on the back page), which shows all the costs and benefits in a perfectly competitive market. At what quantity is economic surplus maximized?

A) Q_1
B) Can't pick any of the other options without more information.
C) Q_2
D) Some quantity higher than Q_2.

11. In Figure 1, what shapes reflect the deadweight loss at the competitive equilibrium?

A) $G + K$
B) $E + F$
C) $C + F$
D) $I + J$

12. In Figure 1, suppose the government required producers to produce Q_2 units (assume the producers comply, that is, they do produce Q_2 units). What is the producer surplus? Hint: In that case, CS would be $A + D + H + I + J$.

A) L
B) $D + E + F + H + L$
C) $L - G - I - J - K$
D) Can't answer without more information.

13. People with a college degree tend to have higher earnings than people without a degree. That is because going to college

A) increases the supply of high-skill workers.
B) tends to improve people's human capital. (⋄)
C) The options labeled (⋄) and (■) are both likely to play a role.
D) is a way of signaling that a person has the traits employers are looking for. (■)

14. Lukas has a car that he values at $5,000. Trip would be willing to pay $7,000 for that car. If Lukas sells the car to Trip for some price over $5,000 and less than $7,000,

A) Both would be better off, but Trip would benefit more than Lukas.
B) Trip should save the extra $2,000 for using Uber or Lyft.
C) Both of them would be better off, but you can't say what the total gains from trade are without knowing the actual price.
D) The increase in total economic surplus as a result of this transaction would be $2,000.

15. I gave Dianna a Spider-Man pen. I bought it at the perfectly competitive market price of $10, and this was my only cost. On her own, Dianna's willingness to pay for this pen was only $5.

A) All of the other options are correct.
B) The gift increased total economic surplus if Dianna valued the sentimental gesture of the gift at more than $5.
C) The gift was inefficient if Dianna would have preferred receiving $10 in cash over receiving the pen.
D) The net effect on total economic surplus is the total benefit Dianna received from the gift minus the $10 cost.

16. In Natestown, two firms operate in the perfectly competitive market for widgets. Both firms have the same constant marginal cost of production. The dirty firm emits 2 units of pollution for each widget it makes. The clean firm emits only 1 unit of pollution for each widget it makes. The Natestown government wants to restrict total Natestown pollution to 10 units, so it issues exactly 10 permits (one permit allows a firm to emit one unit of pollution). Initially, the government gives 5 permits to each firm. After the firms are allowed to freely trade permits,

A) The dirty firm would end up with more permits than the clean firm.

B) The dirty firm will buy permits from the clean firm because it requires more permits to make a single widget.

C) There would be no reason to trade, because each firm wants to make as many units as possible.

D) The clean firm would end up with more permits than the dirty firm.

17. According to the models discussed in class, how does globalization and increased international trade typically affect income inequality *within a high-income country*?

A) It tends to increase income inequality by increasing the demand for products highly-skilled workers make and decreasing demand for products low-skilled workers make.

B) It tends to shrink how much the economy of the rich country produces in goods and services.

C) Increased trade tends to benefit everyone in the country equally.

D) It tends to decrease income inequality because it makes goods cheaper.

18. A local lake is open to everyone for fishing. We expect that

A) In the absence of government intervention, people will bargain with each other (per the Coase Theorem) to ensure that the fishing is kept at an efficient level. (■)

B) The options labeled (⋄) and (■) are both correct.

C) People will fish at the efficient level because they don't want to deplete the lake.

D) The lake will be over-fished because individuals do not account for the negative externality their fishing imposes on others. (⋄)

19. Suppose Simoneland is a very small country that imports widgets from the world market. When the domestic supply curve for widgets in Simoneland shifts right slightly, then

A) The options labeled (⋄) and (■) are both correct.

B) The number of widgets consumed in Simoneland increases. (⋄)

C) Economic surplus in Simoneland increases. (■)

D) The world price for widgets falls substantially.

20. Suppose that each widget produced has a marginal external cost of $5. Then the government could attain zero deadweight loss in the widget market by

A) Imposing a subsidy on demanders of $5 per widget.

B) Banning the production of widgets.

C) None of the other options.

D) Imposing a tax on demanders of $5 per widget.

21. Party time! Your birthday is Friday and you want to invite your four best friends over for a loud party at 8 p.m. However, the law is that you must be quiet after 8 p.m. The party would give you personally $50 worth of benefit, and your four friends will get $10 of benefit *each*. Unfortunately, the party will disrupt your neighbor. He would be willing to pay $75 for you NOT to have the party. Suppose no other people would even notice the party (unless your neighbor calls the cops, who would give you a $1,000 fine). What will happen?

A) If you, your friends, and your neighbor can bargain costlessly, then you will have the party.

B) All of the other options are correct.

C) If you cannot bargain with your neighbor, you will not have the party.
D) It would be efficient to have the party.

Free Response

*Write your response to the following questions on the back of the bubble sheet. Show your work for partial credit. **Box your final answer for each part.***

22. Bob and Linda run a diner. They spend their time making Burgers and baking Cookies. The table below shows their maximum production of each good per 8-hour shift.

Worker	Burgers per shift	Cookies per shift
Bob	25	50
Linda	50	150

a) Who has the absolute advantage in making Burgers? Briefly explain why. Box your answer (for this and each subsequent question).

b) Compute Bob's opportunity cost of making 1 Burger. Show your work.

c) Who has the comparative advantage in baking Cookies? Briefly explain why.

d) Suppose Bob and Linda decide to specialize and trade with one another. What is the range of prices (in terms of cookies) they would agree upon to trade 1 Burger?

23. The table below shows the Marginal Cost (MC) of producing widgets for three different firms. Widgets are sold in a competitive market for **$70 each**. Initially, the government gives each firm exactly **2 permits**. One permit is required to produce one widget. (Assume firms cannot produce partial widgets.)

Widget Number	Firm A's MC	Firm B's MC	Firm C's MC
1st Widget	$15	$20	$25
2nd Widget	$30	$35	$60
3rd Widget	$45	$50	$80

a) Initially, each firm uses its 2 permits to produce 2 widgets. What is the combined total cost of production for these 6 widgets?

b) Suppose no permits have been traded yet. What is Firm A's **maximum** willingness-to-pay (WTP) for a 3rd permit, and what is Firm C's **minimum** willingness-to-accept (WTA) to sell one of its 2 permits?

c) Assume transaction costs are zero and firms are allowed to trade permits. After all mutually beneficial trades have occurred, what is the new total cost of production for the 6 widgets?

Do not turn this exam over until instructed to.

Be sure to put your name and exam version on BOTH sides of your response sheet.

For the multiple choice problems, choose the best possible option. Use the model we have developed in the course.

Good luck!

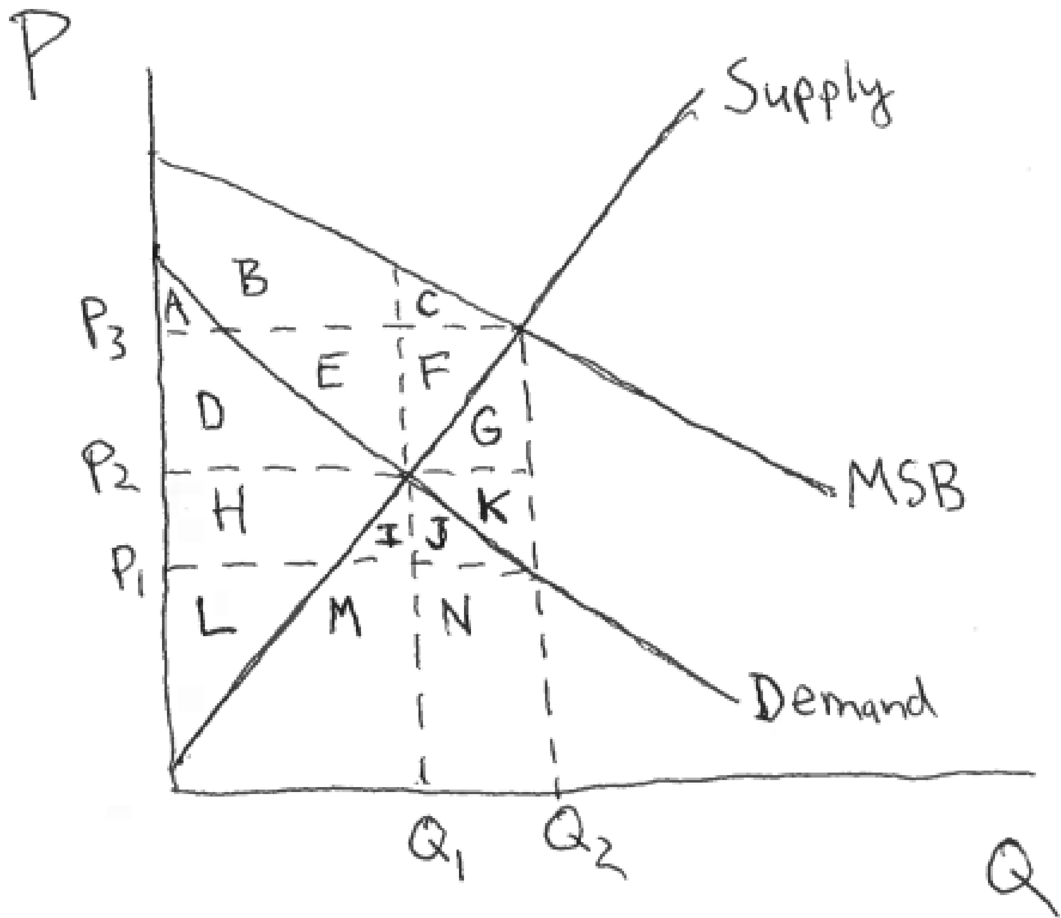

Figure 1